

Line Number	Identifier	Error Type	Error	Error Correction
15	Global Variable (totalBooks)			
17	Function Prototype (addBook)			
24	Variable (choice)	Syntax Error	Missing semicolon after declaration	int choice;
58	Switch Statement	Syntax Error	Missing colon after default	default:
92	Loop Condition	Runtime Error	Loop accesses one element beyond array	Use i < totalBooks
154	Expression	Runtime Error	Possible division by zero when totalBooks is 0	Check totalBooks before division
157	Variable (totalBook)	Scope Error	Undeclared variable	Use totalBooks
160	Function Call (saveLibrary)	Linker Error	Function used before declaration	Add prototype
162	Function Call (calculateFine)	Linker Error	Function not declared/defined	Declare and define function
169	Variable (bookId)	Semantic Error	String assigned to int	b.bookId = 101;
175	Variable (price)	Semantic Error	String assigned to float	b.price = 500.0;
186	String Comparison	Logical Error	Arrays compared using ==	Use strcmp()
197	Structure Array	Semantic Error	Cannot assign NULL to structure object	Use flag or shift elements
215	Function Call (calculateAveragePrice)	Linker Error	Function not declared/defined	Declare and define function
228	File Operation	Runtime Error	File opened in read mode then written	Use fopen(...,"w")
264	Pointer (ptr)	Runtime Error	NULL pointer dereference	Assign valid address
268	Array Access	Runtime Error	Index 10 out of bounds	Use index 0-4
273	Expression	Runtime Error	Division by zero	Ensure b != 0
282	String Operation	Runtime Error	Buffer overflow in strcpy	Increase array size or limit input
291	Memory Allocation	Runtime Error	malloc(5) allocates insufficient bytes	Use malloc(5 * sizeof(int))
299	Pointer Usage	Runtime Error	Accessing freed memory	Do not use ptr after free
301	Memory Management	Runtime Error	Double free detected	Free pointer only once